Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 0.333... (the 3 repeats forever) a rational number? Student says: No

Decide if the decimal repeats or ends.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $\sqrt{2}$ a rational number? Student says: Yes

Decide if the decimal repeats or ends.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 0.25 a rational number? Student says: No

Write the decimal as a fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is π a rational number? Student says: Yes

Decide if the decimal repeats or ends.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.777… (the 7 repeats forever) as a fraction. Student's answer: 7/10

Let x = 0.777…, then multiply by 10 and subtract.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

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Find the Error

8.NS.1

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

the decimal 0.1010010001... (the number of zeros grows, with no repeating pattern) rational? Student says: Y

Look for a repeating pattern.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Between which two whole numbers is $\sqrt{50}$? Student's answer: between 5 and 6

Find the perfect squares near 50.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Locate $\sqrt{10}$ on a number line. Student says: It is closer to 4.

Find the nearby whole numbers and compare.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare $\sqrt{7}$ and 2.6. Student says: $\sqrt{7}$ is less than 2.6.

Square each value to compare.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 3.14 and $\sqrt{9.8}$. Student says: $\sqrt{9.8}$ is greater.

Estimate the square root.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Approximate $\sqrt{48}$ to the nearest tenth. Student's answer: 6.8

Estimate and check.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Approximate π to the nearest hundredth. Student's answer: 3.1

Look at the thousandths digit.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $2^3 \times 2^2$. Student's answer: 2

Use the rule for multiplying powers with the same base.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 5. Student's answer: 0

Use the rule for zero exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 10^2. Student's answer: -100

Use the rule for negative exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $(3^2)^3$. Student's answer: 3

Use the rule for a power of a power.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 4^1. Student's answer: -4

Use the rule for negative exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $x \div x^2$. Student's answer: x

Use the rule for dividing powers with the same base.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find $\sqrt{49}$. Student's answer: 24.5

Find the number that times itself is 49.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 = 36$ for the positive value of x. Student's answer: $x = 3$

Take the square root of both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find 27. Student's answer: 9

Find the number that times itself three times is 27.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 = 81$. Student's answer: $x = 9$

Find all solutions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find $\sqrt{0.25}$. Student's answer: 0.125

Find the number that times itself is 0.25.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find (-8). Student's answer: no real solution

Think about the sign of a cube.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 4.5×10^3 in standard form. Student's answer: 450

Move the decimal point.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.00052 in scientific notation. Student writes: 5.2×10

Move the decimal point to get a number between 1 and 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is greater: 3×10 or 5×10^3? Student's answer: 5×10^3

Compare the powers of 10 first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(2 \times 10^3)(3 \times 10^2)$. Student's answer: 5×10

Multiply the coefficients and add the exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 6,200,000 in scientific notation. Student writes: 6.2×10

Count the places the decimal moves.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The distance to a star is 9.46×10^{12} km. Is this number written in standard form? Student says: Yes

Check the form.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A city has 2.4×10 people. How many people are in 3 cities of the same size? Student's answer: 7.2×10

Multiply the coefficient, keep the power of 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide $(8 \times 10) \div (2 \times 10^2)$. Student's answer: 4×10^3

Divide the coefficients and subtract the exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(6 \times 10^3)(5 \times 10^2)$. Student's answer: 3×10

Multiply the coefficients and add the exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Light travels 3×10 km each second. How far does it travel in 2×10^2 seconds? Student's answer: 6×10 km

Multiply the rate by the time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.0045 in scientific notation. Student writes: 4.5×10^3

Move the decimal point to the right of the first nonzero digit.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 12 × 10 in scientific notation? Student says: Yes

Check that the coefficient is between 1 and 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (0, 0) and (2, 6). What is its slope? Student's answer: 1/3

Use rise over run.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line rises 4 units for every 2 units across. What is its slope? Student's answer: 1/2

Use rise over run.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (1, 5) and (3, 1). What is its slope? Student's answer: 2

Subtract the y-values and the x-values in the same order.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which line is steeper: slope 3 or slope 1/2? Student says: slope 1/2

Compare the sizes of the slopes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line with slope 0.5 goes through the origin. What is y when x = 10? Student's answer: 20

Multiply the slope by x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which line is increasing: slope -4 or slope 2? Student's answer: slope -4

Think about the sign of the slope.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (0, 0) and (1, 4). Write its equation. Student writes: $y = x + 4$

Find the slope, then the y-intercept.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line has a slope of 2 and a y-intercept of 3. Write its equation. Student writes: $y = 3x + 2$

Use $y = mx + b$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (0, -2) and (2, 2). Write its equation. Student writes: $y = -2x + 2$

Find the slope, then the y-intercept.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is the point (3, 11) on the line $y = 2x + 5$? Student says: No

Substitute 3 for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (1, 5) and (2, 8). Write its equation. Student writes: $y = 3x + 5$

Find the slope, then use a point to find b.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line has a slope of -1/2 and a y-intercept of 4. Write its equation. Student writes: $y = 1/2x + 4$

Use $y = mx + b$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3(x + 2) = 2x + 9$. Student's answer: $x = 15$

Distribute, then collect the x terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2x + 3 = 2x + 5$. Student's answer: $x = 1$

Subtract 2x from both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $4(x + 1) = 4x + 4$. Student's answer: $x = 0$

Distribute, then compare both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $5 - (x - 2) = 2$. Student's answer: $x = -5$

Distribute the negative sign to both terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x/3 + 1 = 4$. Student's answer: $x = 3$

Undo the addition, then multiply by 3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2(x - 3) = 10$. Student's answer: $x = 6.5$

Distribute, then solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = x + 1$ and $y = 3x - 3$. Student's answer: (1, 2)

Set the two expressions equal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $x + y = 10$ and $x - y = 4$. Student's answer: $(3, 7)$

Add the equations to eliminate y.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = 2x + 1$ and $y = 2x + 5$. Student's answer: (1, 3)

Compare the two equations.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is (2, 3) a solution of the system $y = x + 1$ and $y = 3x - 3$? Student says: No

Substitute into both equations.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The lines $y = 2x$ and $y = -x + 6$ cross at a point. What is the point? Student's answer: (4, 2)

Set the two expressions equal and solve for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

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Find the Error

8.EE.8

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

5 + 2x dollars and Plan B costs 3x dollars. At what number of items do the plans cost the same? Student's

Set the two costs equal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is the relation {(1, 2), (1, 3), (2, 4)} a function? Student says: Yes

Check whether any input has two outputs.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is y = 2x a function? Student says: No

Check that each input gives one output.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A graph crosses a vertical line in two places. Is the graph a function? Student says: Yes

Use the vertical line test.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table maps input 3 to output 9 and input 4 to output 9. Is this a function? Student says: No

Check each input.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A relation maps input 5 to outputs 6 and 7. Is it a function? Student says: Yes

Check each input.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $y = |x|$ a function? Student says: No

Check that each input gives one output.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

rate of change of 3 from a table. Function B is y = 2x + 3. Which has the greater rate of change? Student's a

Compare the rates of change.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

y = 3x + 1. Function B has the points (0, 2) and (1, 5). Which has the greater starting value? Student's answe

Find the starting value of each.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s y = 5x. Function B has the points (0, 0) and (1, 4). Which has the greater rate of change? Student's answer

Compare the slopes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $y = 2x^2$ a linear function? Student says: Yes

Check whether the rate of change is constant.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has the points (1, 3), (2, 6), and (3, 10). Is it linear? Student says: Yes

Check whether the rate of change is constant.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

tance d = 40t. Car B has the points (1, 30) and (2, 60) for time and distance. Which car is faster? Student's

Compare the rates of change.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find y when $y = x^2$ and $x = 3$. Student's answer: 6

Square the input.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which function is nonlinear: $y = 4x + 1$ or $y = x^2$? Student's answer: $y = 4x + 1$

Look for the variable raised to a power.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The table has x = 1, 2, 3 and y = 1, 4, 9. Is the relationship linear? Student says: Yes

Check whether the rate of change is constant.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The area of a square is $A = s^2$, where s is the side length. Is this relationship linear? Student says: Yes

Check for a squared variable.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $y = 1/x$ a linear function? Student says: Yes

Check the rate of change.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which way does the graph of $y = x^2$ open? Student's answer: downward

Think about the sign in front of x^2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has the points (0, 3) and (2, 7). Write the linear function. Student writes: $y = 3x + 2$

Find the rate of change and the initial value.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the initial value of y = 5x - 4? Student's answer: 5

Find the value of y when x = 0.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

k has 10 gallons and loses 2 gallons each hour. Write the function. Student's answer: starting value 10 and ra

Identify the starting amount and the rate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A savings account starts with $50 and gains $15 each week. What is the initial value? Student's answer: 15

Find the starting amount.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line passes through (0, 6) and (3, 0). What is the rate of change? Student's answer: 2

Use (change in y) ÷ (change in x).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the rate of change for the points (1, 4) and (3, 10). Student's answer: 6

Use (change in y) ÷ (change in x).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

oh starts high on the left and falls steadily to the right. Is it increasing or decreasing? Student's answer: incre

Look at the direction of the graph from left to right.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

at for a while and then rises steadily. Which description fits the flat part? Student's answer: increasing at a c

Think about what a flat graph means.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

re graph rises from 40 to 80 and then falls to 50. When is the temperature decreasing? Student's answer: f

Find the part of the graph going down.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A graph is a curve that gets steeper as x increases. Student says: The rate of change is constant.

Think about whether the steepness changes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ws distance rising to 12 miles at 2 hours, then staying at 12 for 1 hour. What happens in the last hour? Studen

Look at the shape of the graph in that hour.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A graph starts at the point (0, 5). What is the initial value? Student's answer: 0

Find y when x = 0.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Translate the point (2, 3) right 4 units. Student's answer: (2, 7)

Move the x-value for a horizontal shift.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Reflect the point (2, 3) across the x-axis. Student's answer: (-2, 3)

Think about what changes in a reflection across the x-axis.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Reflect the point (2, 3) across the y-axis. Student's answer: (2, -3)

Think about what changes in a reflection across the y-axis.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Rotate the point (1, 0) 90° counterclockwise about the origin. Student's answer: (1, 0)

Turn the point a quarter turn counterclockwise.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Dilate the point (2, 3) by a scale factor of 2 from the origin. Student's answer: (2, 6)

Multiply both coordinates by the scale factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Translate the point (5, 5) left 3 units and down 2 units. Student's answer: (8, 7)

Subtract from x and y.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are two rectangles that are both 3 by 5 congruent, with one rotated? Student says: No

Check whether the size and shape match.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are triangles with sides 3, 4, 5 and 4, 5, 6 congruent? Student says: Yes

Compare the side lengths.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle is reflected across a line to make a new triangle. Are they congruent? Student says: No

Think about what a reflection does.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

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Find the Error

8.G.2

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

nsformation maps a figure to a congruent figure: translation, reflection, rotation, or dilation? Student's answ

Think about which ones keep size.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

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Find the Error

8.G.2

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is a square with side 4 congruent to a square with side 8? Student says: Yes

Compare the side lengths.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two triangles have the same three angles but different sizes. Are they congruent? Student says: Yes

Think about size as well as angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Dilate the point (3, -1) by a scale factor of 2 from the origin. Student's answer: (5, 1)

Multiply both coordinates by 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Dilate the point (6, 10) by a scale factor of 1/2 from the origin. Student's answer: (3, 4)

Multiply both coordinates by 1/2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

pe has a side of 2 units. It is dilated by a scale factor of 3. What is the new side length? Student's answer: 5

Multiply the side by the scale factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Dilate the point (2, 3) by a scale factor of 2 from the center (1, 1). Student's answer: (4, 6)

Find the distance from the center and double it.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A figure is dilated by a scale factor of 0.5. Student says: The new figure is larger.

Compare the scale factor to 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A figure is dilated by a scale factor of 3. Student says: The area becomes 3 times as large.

Think about how area scales.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Triangles have sides 3, 4, 5 and 6, 8, 10. Are they similar? Student says: No

Check whether the sides have the same ratio.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

es a side of 6 in a similar triangle. The matching side is 10 in the first triangle. What is the matching side in t

Set up a proportion.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Rectangles are 2 by 3 and 4 by 6. Are they similar? Student says: No

Compare the ratios of the sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are two triangles similar if two pairs of angles match? Student says: No, you need the sides too.

Think about angle-angle similarity.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Rectangles are 3 by 5 and 6 by 9. Are they similar? Student says: Yes

Compare the ratios of the sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

with sides 3 and 9 are similar. What is the scale factor from the small square to the large one? Student's ar

Divide the large side by the small side.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has angles 50° and 60°. What is the third angle? Student's answer: 80°

Angles in a triangle add to 180°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

parallel lines are cut by a transversal. One alternate interior angle is 65°. What is the other? Student's answe

Think about alternate interior angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ıre cut by a transversal. A corresponding angle is 40°. What is the matching corresponding angle? Student'

Think about corresponding angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

n exterior angle of 120° and one remote interior angle of 70°. What is the other remote interior angle? Stude

An exterior angle equals the sum of the two remote angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has angles 40°, x, and 100°. What is x? Student's answer: 140°

Angles in a triangle add to 180°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s are cut by a transversal. Same-side interior angles include one angle of 110°. What is the other? Student's

Same-side interior angles add to 180°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has legs 3 and 4. What is the hypotenuse? Student's answer: 7

Use $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has legs 6 and 8. What is the hypotenuse? Student's answer: 14

Use $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has a hypotenuse of 13 and a leg of 5. What is the other leg? Student's answer: 8

Use $a^2 + b^2 = c^2$, solving for the missing leg.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has a hypotenuse of 15 and a leg of 9. What is the other leg? Student's answer: 6

Use $a^2 + b^2 = c^2$, solving for the missing leg.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is a triangle with sides 5, 12, and 13 a right triangle? Student says: No

Check whether $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A square has a side of 4. What is the length of its diagonal? Student's answer: 8

Use the Pythagorean theorem with the two sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangular floor is 12 feet by 5 feet. What is the length of the diagonal? Student's answer: 17 feet

Use $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ler 10 feet long leans on a wall with its base 6 feet from the wall. How high does it reach? Student's answer:

Use $a^2 + b^2 = c^2$, solving for the height.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box is 3 by 4 by 12 units. What is the length of its space diagonal? Student's answer: 19

Use the 3D version of the Pythagorean theorem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has a hypotenuse of 17 and a leg of 8. What is the other leg? Student's answer: 9

Solve for the missing leg.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

eet from a point on the ground. A wire runs from the top of the pole to that point. How long is the wire? Stud

Use $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is a triangle with sides 6, 8, and 10 a right triangle? Student says: No

Check whether $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (0, 0) and (6, 8). Student's answer: 14

Use the distance formula or the Pythagorean theorem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (1, 1) and (4, 5). Student's answer: 7

Find the horizontal and vertical distances.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (-2, 3) and (1, -1). Student's answer: 7

Find the horizontal and vertical distances.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (2, 2) and (2, 7). Student's answer: 7

Find the vertical distance.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has vertices (0, 0), (3, 0), and (0, 4). What is the length of its hypotenuse? Student's answer: 7

Find the distance between the two vertices that are not on the same axis.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (0, 0) and (5, 12). Student's answer: 17

Use the distance formula.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a cylinder with radius 2 and height 3. Student's answer: 6π

Use $V = \pi r^2 h$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a cone with radius 3 and height 4. Student's answer: 36π

Use $V = (1/3)\pi r^2 h$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a sphere with radius 3. Student's answer: 12π

Use $V = (4/3)\pi r^3$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cone has radius 2 and height 6. What is its volume? Student's answer: 24π

Use $V = (1/3)\pi r^2 h$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sphere has a diameter of 6. What is its volume? Student's answer: 288π

Find the radius first, then use $V = (4/3)\pi r^3$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cylinder has a volume of 50π and a radius of 5. What is its height? Student's answer: 10

Solve $V = \pi r^2 h$ for h.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.